FastSpeech 2
English and Telugu

Technical report | Implementation and execution status | 7 November 2026

Execution status and evidence

This project implements independent English and Telugu text-to-speech pipelines based on the supplied FastSpeech2-master source. The code supports corpus preparation, learned duration alignment, acoustic-model training, synthesis and variance visualization. At the latest execution evidence supplied on 7 November 2026, English dataset export had completed and acoustic feature extraction was still running. Full language-model training and evaluation had not been demonstrated. This report documents the implemented method, observed execution progress and remaining experimental work; it does not claim that all assignment deliverables are complete.

Deliverable	Status supported by available evidence
English data export	Completed: exporter printed the local manifest path
English feature extraction	In progress: file count increased from 698 to 726; later screenshots show continued feature extraction
Telugu data preparation	No completed run evidenced
Implementation checks	Seven synthetic contract/integration tests passed in the package-authoring environment
English training	Pending: no real checkpoint or measured loss curve available for this report
English synthesis	Pending: no real test audio or variance plots available for this report
Telugu training	Pending: no real checkpoint or measured loss curve available for this report
Telugu synthesis	Pending: no real test audio or variance plots available for this report

Scope of evidence: execution status is based on the provided terminal outputs and screenshots, together with inspected project code. The Windows project directory was not independently inspected. Reported file counts are progress observations, not a measurement of selected audio hours or a percentage complete. Results produced after this snapshot must be incorporated in an updated report.

Architecture and objective

Unicode token IDs enter the upstream FFT encoder. Speaker/group embeddings condition the encoder states. Duration, pitch and energy predictors produce token-level values; pitch and energy embeddings condition the states and a length regulator expands them to mel frames. The upstream FFT decoder, linear mel projection and five-layer convolutional PostNet predict 80-band log-power mel spectra. Griffin-Lim reconstructs audio from these predicted spectra.

The training objective is mel L1 + postnet mel L1 + duration MSE on log(1 + frame count) + normalized pitch MSE + normalized energy MSE. All losses mask padded positions. Ground-truth variances condition training and validation; free-running inference uses predicted variances. Validation loss alone cannot establish intelligibility or pronunciation quality.

Implementation and language adaptations

Code provenance

Reference: ming024/FastSpeech2, supplied archive commit d4e79eb52e8b01d24703b2dfc0385544092958f3. The MIT license and upstream README are retained. The FFT layers, variance predictors and length regulator are used from that implementation. CHANGES.md lists each modified module. This is a FastSpeech 2 adaptation, not a paper-exact reproduction.

Component	Implemented choice
Transformer	4 encoder / 6 decoder FFT layers; width 256; 2 heads; convolution width 1024; kernels 9 and 1
Variance module	Width 256; kernel 3; dropout 0.5; 256 linear bins; token-level pitch/energy
Training	Independent language runs; AdamW LR 0.0001; batch 8; gradient norm cap 1; seed 42; initial budget 150 epochs
Alignment	Two-layer bidirectional LSTM CTC; hidden 192 per direction; Adam LR 0.001; initial budget 40 epochs
Acoustics	22,050 Hz mono; FFT/window 1024; hop 256; 80 mel bands; 0-8000 Hz; pYIN F0 50-600 Hz

English uses a fixed ASCII-plus-punctuation inventory. Telugu adds the full U+0C00-U+0C7F block and retains combining marks, virama, and joiners. These are Unicode code-point tokens, not phonemes or complete orthographic syllables. Unsupported text raises an error and is logged during selection; it is never silently stripped. Both language frontends apply NFC and lowercase only. Original test text is retained separately from model input.

No language-specific numeric expansion, heteronym resolver or code-switch transliteration is asserted. Literal test anomalies are retained. Telugu character durations can partition dependent marks artificially; this limitation motivates future syllable/phoneme input with a vetted language aligner.

Implementation fixes include device-local length tensors, corrected energy control, padded-duration masking, duration scaling before rounding, and explicit failure for empty or excessive synthesized sequences. Separate configs, vocabularies, checkpoints and outputs prevent accidental cross-language training.

Execution environment and progress evidence

Item	Observed value / status
Operating environment	Windows PowerShell; project virtual environment; Python 3.11 indicated by the supplied traceback
Installed PyTorch	2.15.0+cpu; torch.version.cuda returned None; CUDA availability returned False
Discrete GPU	NVIDIA GeForce GTX 1650; 4096 MiB memory
NVIDIA driver	592.82; nvidia-smi displayed CUDA Version 13.1
GPU snapshot	110 MiB used and 2% GPU utilization at 20:39:34 IST on 7 November 2026; one observation, not a benchmark
English data	Export completed and manifest path printed; feature extraction subsequently produced successive features english_... messages
Measured progress	Two supplied file-count checks showed 698 and 726 .npz feature files; no elapsed interval was provided

The CPU-only PyTorch build explains why the detected NVIDIA GPU was unavailable to the training code at the time of inspection. The CUDA version displayed by nvidia-smi is driver capability information; it is not proof that the Python environment contains a CUDA-enabled PyTorch build. Feature extraction itself uses CPU audio-processing routines, including pYIN, so changing PyTorch does not directly accelerate that stage.

A CUDA-enabled PyTorch 2.7.1 build for CUDA 12.6 and smaller batches were recommended after preprocessing: alignment batch size 2 and acoustic-model batch size 1. These are proposed settings, not confirmed changes or measured performance improvements. The supplied configurations retain batch size 8 unless edited locally. Long sequences may still exceed the 4 GB GPU memory budget.

Observed issue and response: the initial dataset request failed because Svarah required authentication. A subsequent export completed, showing that access was resolved sufficiently for that stage. The Windows symlink warning did not prevent export. Slow but increasing feature counts showed ongoing work; no throughput or completion-time estimate can be computed from two counts without timestamps.

Software verification evidence

The package verification record documents seven passing tests in a separate CPU authoring environment: CTC backpropagation; repeated-token Viterbi alignment; train-only feature statistics; Telugu text preservation; split duplicate protection; waveform feature extraction; and two-language synthetic train/resume/synthesis export. The final integration test used tiny models and temporary fixtures. Its waveform-export check fixed temporary duration biases; it does not demonstrate learned speech.

The authoring test environment used Python 3.12.14, PyTorch 2.15.0+cpu, librosa 0.11.0 and SoundFile 0.13.1. These tests do not establish full-corpus convergence, Windows GPU compatibility, pronunciation quality or equivalence to the research paper. Real GPU training and end-to-end dataset validation remain to be completed.

Dataset preparation and duration supervision

The exact assignment dataset identifiers are ai4bharat/Svarah and SPRINGLab/IndicTTS_Telugu [3, 4]. Svarah exposes an ASR test split; it is repurposed for this assignment only. Its Hugging Face schema lacks verified speaker IDs, so English uses a pooled voice. The Telugu card describes two speakers; its gender labels identify those two voices for this corpus. English pooled voice variability is an important confound.

Decoded recording hashes remove exact duplicate audio. Normalized identical transcripts form groups that remain wholly in one partition; both Telugu voices can therefore be retained without identical-text leakage between partitions. Groups are shuffled with seed 42, then accumulated to at least five training hours and one validation hour. Excess greater than 60 seconds fails. The method permits speaker overlap and evaluates held-out utterances, not unseen speakers. Rejected files, chosen IDs, waveform hashes and actual durations are recorded.

Waveforms are mixed to mono and resampled without silence trimming, preserving the timing axis. Log power mel features, pYIN F0 and RMS energy share the same STFT frame count. An independent CTC acoustic aligner learns transcript alignment on the training partition only. Monotonic Viterbi includes blank and repeated-label transitions. Every mel frame is assigned to a token; leading blanks attach to the first token and interior blanks to the preceding token. Duration sums must equal mel length.

The best aligner is chosen by greedy CER on a fixed training subset, not validation. CER above 0.35 aborts export. A low subset CER does not guarantee good alignments: inspect full per-utterance CER, log-probability, long-duration outliers and sample boundaries. This procedure is learned alignment supervision, not equal-duration pseudo-labels.

Token F0 averages voiced frames, with wholly unvoiced tokens assigned zero. Energy averages token RMS. Normalization means and standard deviations are computed solely on training token targets, including zero F0 tokens. Validation reuses those statistics. Predictions are inverse-transformed for plots; negative F0/energy are clipped for display and the raw values are retained in JSON.

Language	Measured selected hours	Evidence
English	NOT MEASURED	Requires full data preparation
Telugu	NOT MEASURED	Requires full data preparation

Training and validation analysis

English

PENDING. No completed real-data training run or loss history was supplied. No claim of convergence, instability, overfitting or comparative performance is supported. After training, inspect component curves as well as total loss. A growing train-validation gap may indicate overfitting; abrupt joint spikes may indicate corrupted targets or optimization instability. These are interpretation criteria, not observations.

Telugu

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Checkpoint selection uses minimum validation total. last.pt supports resume and best.pt supports synthesis. Optimizer, completed epoch, RNG states, configuration, source split identity, vocabulary and statistics are saved or retained alongside the checkpoints. The data/architecture fingerprint rejects incompatible checkpoints.

Test sentence analysis and listening protocol

All five utterances per language are defined in the UTF-8 test_sentences.json files. The a/b pairs are separate outputs. Text from the rendered PDF is preserved, including the unusual internal spaces; glyph-layout artifacts from raw PDF extraction are not treated as linguistic edits. The JSON files are the authoritative transcript records in this project.

ID	Question to evaluate after synthesis
E1	Does the long word remain intelligible, with plausible segment timing and no collapse?
E2a	Inspect noun conduct: first-syllable stress and its duration/F0/energy cues; compare with E2b. Correct realization is unmeasured.
E2b	Inspect verb conduct: second-syllable stress and contextual rhythm; compare with E2a. Correct realization is unmeasured.
E3	Does the literal split rhyth m create an unnatural break, omission or spelling-like output?
E4	Check time/date reading and the code-switched andaroo. No explicit normalization model is included.
T1	Evaluate the long technical loanword, dependent vowels, consonant clusters and overall articulation.
T2a	Inspect pattu in the grip context: consonant timing, stress/prominence and phrase contour. Semantic differences need not change phonemes.
T2b	Inspect pattu in the silk-sari context and compare with T2a. Evaluate phrase prosody without assuming a mandatory phonemic contrast.
T3	Evaluate whether the deliberate split in the word for forest produces a pause or pronunciation error.
T4	Check numerals, ordinal suffix, time expression, phrasing and sentence ending.

Recorded listening/variance observations (NOT EVALUATED means no empirical claim):

All ten sentences: pronunciation, quality, duration, pitch and energy are NOT EVALUATED. The listening_notes.json file provides an explicit field for every required observation.

Variance evidence, comparison and reproducibility

English variance evidence

NOT GENERATED. Duration, pitch and energy plots require actual inference. No observed peaks, pauses, stress patterns or speech quality are inferred in their absence.

Telugu variance evidence

NOT GENERATED. Duration, pitch and energy plots require actual inference. No observed peaks, pauses, stress patterns or speech quality are inferred in their absence.

Plot x-axes are Unicode token indices, with exact text mappings in adjacent JSON. Token means cannot show within-token F0 movement. Zero F0 denotes an unvoiced target convention, not a measured phonation label for synthesized audio. Duration sums give the mel-frame count; centered-STFT reconstruction can differ by roughly one hop in waveform duration. Compare inverse-transformed units within matched voices, and account for voice F0 ranges before comparing languages.

No English-versus-Telugu winner can be claimed without the runs. Expected confounds include multi-speaker pooled English versus two conditioned Telugu voices, noisier ASR-source recordings versus studio TTS recordings, alignment quality, character segmentation, text normalization and the reconstruction method. Griffin-Lim artifacts must be separated from model pronunciation failures through mel/variance inspection and listening. Improvements include verified speaker labels, vetted phoneme alignments, syllable-aware Telugu input, contextual numeric normalization and a matching trained neural vocoder.

Reproducible commands

python English/preprocess.py --manifest data/raw/English/manifest.json

python -m fs2.alignment --config English/config.yaml --device auto

python English/train.py --device auto; then python English/inference.py --device auto

python Telugu/preprocess.py --manifest data/raw/Telugu/manifest.json

python -m fs2.alignment --config Telugu/config.yaml --device auto

python Telugu/train.py --device auto; then python Telugu/inference.py --device auto

Run python scripts/smoke_test.py first. See README.md for virtual environments, CPU/CUDA installation, dataset export, licenses, resume, output paths and GitHub submission. After listening, edit Report/listening_notes.json and run python scripts/build_report.py followed by python scripts/check_submission.py. Software ranges are in requirements.txt; the authoring test environment is recorded in TESTING.md.

References

[1] Ren et al. FastSpeech 2: Fast and High-Quality End-to-End Text to Speech. https://arxiv.org/abs/2006.04558

[2] ming024/FastSpeech2. https://github.com/ming024/FastSpeech2 (supplied source archive)

[3] AI4Bharat Svarah: https://huggingface.co/datasets/ai4bharat/Svarah and https://github.com/AI4Bharat/Svarah

[4] SPRINGLab IndicTTS Telugu: https://huggingface.co/datasets/SPRINGLab/IndicTTS_Telugu

[5] Original IndicTTS license: https://www.iitm.ac.in/donlab/indictts/downloads/license.pdf

Appendix: prescribed test sentences

The following tables are reproduced directly from the supplied assignment PDF, pages 1 and 2. Using the rendered source preserves the Telugu glyph shaping and the deliberately unusual internal spaces. E2 and T2 each contain two utterances, giving five utterances per language. These are input texts, not synthesized outputs.

English Test Sentences

ID	Test sentence
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.
E2	a) The orchestra's conduct during the ceremony was praised by the audience. b) The researchers will conduct an experiment to evaluate the proposed method.
E3	The rhyth m of the orchestra changed unexpectedly.
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

Telugu Test Sentences

ID	Test sentence
T1	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.
T2	a) తాడుపై అతని పట్టు చాలా బలంగా ఉంది. b) ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.
T3	పులి అడ విలో నిశ్శబ్దంగా నడిచింది.
T4	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

Project input files: English/test_sentences.json and Telugu/test_sentences.json. Inference records each original text, its NFC/lowercase model input and token index mapping. No automatic correction of the broken English word or Telugu spacing is applied. The a/b cases are exported under distinct IDs: E2a, E2b, T2a and T2b.

Remaining evidence required for a completed evaluation

Both trained language checkpoints; final train/validation split totals and manifests; loss histories and curves; ten synthesized WAV files; thirty duration/pitch/energy plots; sentence-level listening observations; a supported cross-language comparison; and the accessible project repository or checkpoint download locations. No repository URL is asserted in this report because one was not supplied.